

WP4 — PROVING THE PROPOSED CONCEPT



A Statistical Approach to Monitor Quantisation in Neural Network Training

Seminar No. 4 — implementation and concept validation



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Project 09I03-03-V04-00562 · SAMQ-NN

Agenda



Summary of completed project tasks and outputs



Concept overview: the statistical similarity index



Implementation — core code and training strategy



Experimental validation of the concept (WP4)



Findings, limitations and next steps



Discussion

Seminar objectives



Present the concept — Monitoring quantisation with statistical metrics (KL, JS divergence).



Explain the core code — Characterisation, PTQ, genuine QAT and per-channel quantisation.



Present the strategy — A new training strategy driven by the similarity index.



Substantiate validation — Experiments across datasets and architectures (WP4).

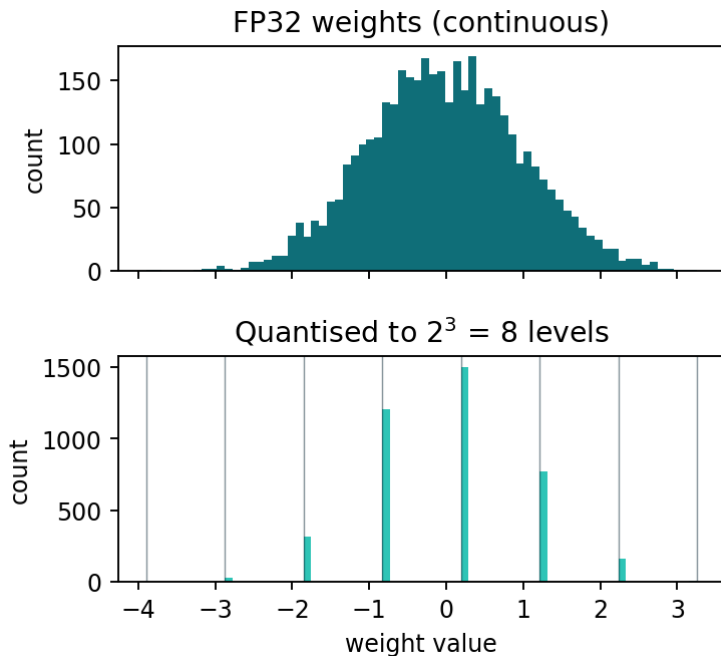


Open the discussion — Results and further direction — final report D17.

Why monitor quantisation?

Quantisation reduces the network's weights to fewer bits — saving memory and compute, but it can degrade accuracy.

- Weights are snapped onto a grid of 2^b levels instead of continuous values.
- Project question: can we statistically track the impact of quantisation on the weight distribution during training?
- Goal: a cheap, interpretable indicator of degradation.

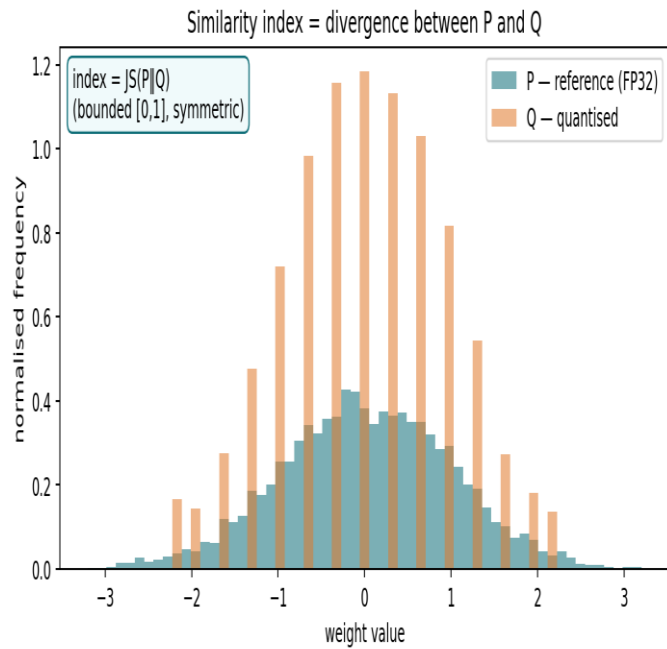


Statistical similarity index

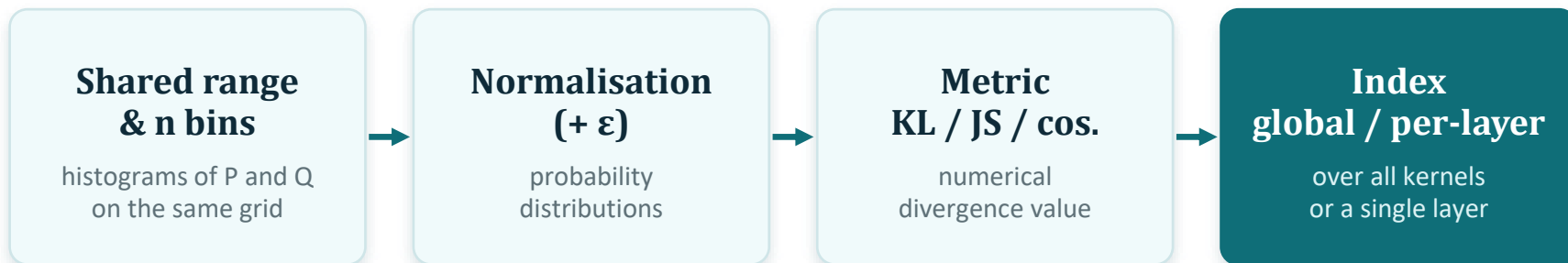
Index = divergence between the weight distribution of the reference (FP32) model **P** and the quantised model **Q**.

- **KL divergence** — sensitive, unbounded — detects information loss.
- **JS divergence** — bounded [0,1], symmetric — suited to thresholding (default).
- **Cosine similarity** — rough comparison of the overall orientation of the distributions.

“A thermometer of weight-distribution shape.”



How the index is computed



The **global index** tracks the whole model; the **per-layer index** reveals which layer is most sensitive to quantisation.

Implementation: core of the source code



Weight characterisation

Moments, Shannon/differential entropy, histograms with consistent bin edges.



Post-training quantisation

`quantize_array` — snapping weights onto a grid of 2^b levels (per-channel).



Genuine QAT (STE)

`QuantConv2D` / `QuantDense` — forward-quantised weights, gradient via STE.



Per-channel quantisation

Own min-max scale per output channel — robust even for deep networks.

New training strategy (SMT)



SimilarityMonitor

index after every epoch — live



IndexEarlyStopping

stops when index and accuracy
stabilise



auto_select_bitwidth

lowest bit-width with JS $\leq$
threshold

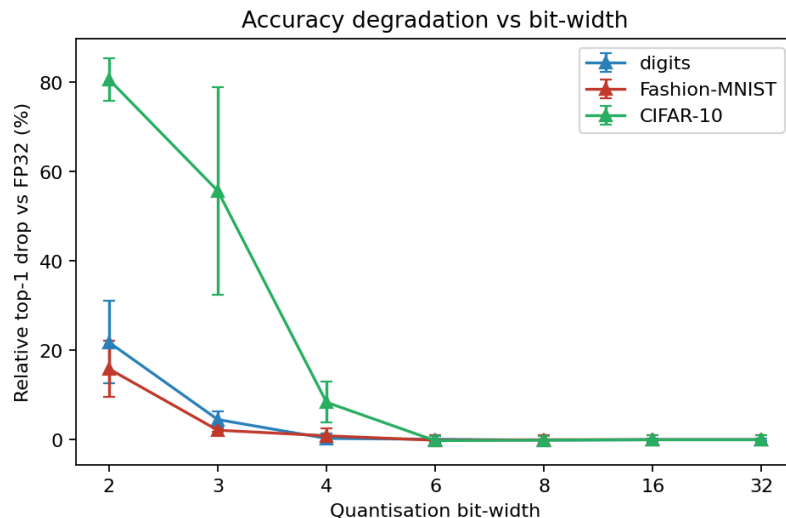
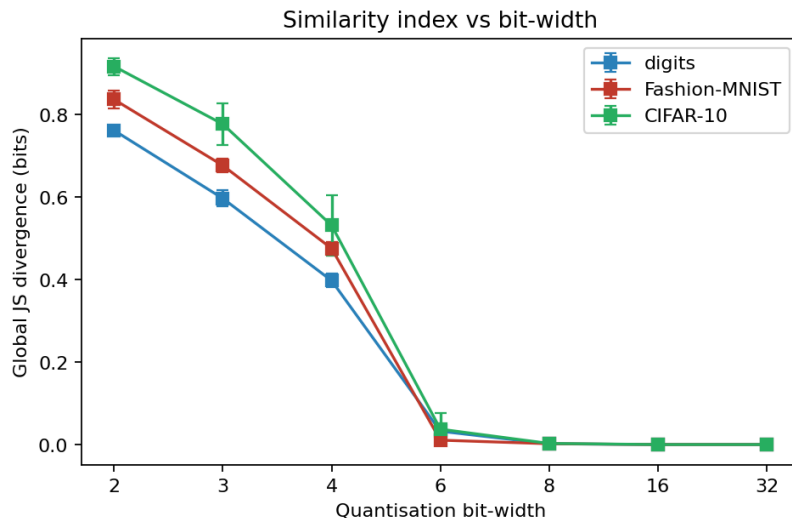


Genuine QAT driven by the similarity index

three components combined into a single training procedure

Sensitivity to bit-width

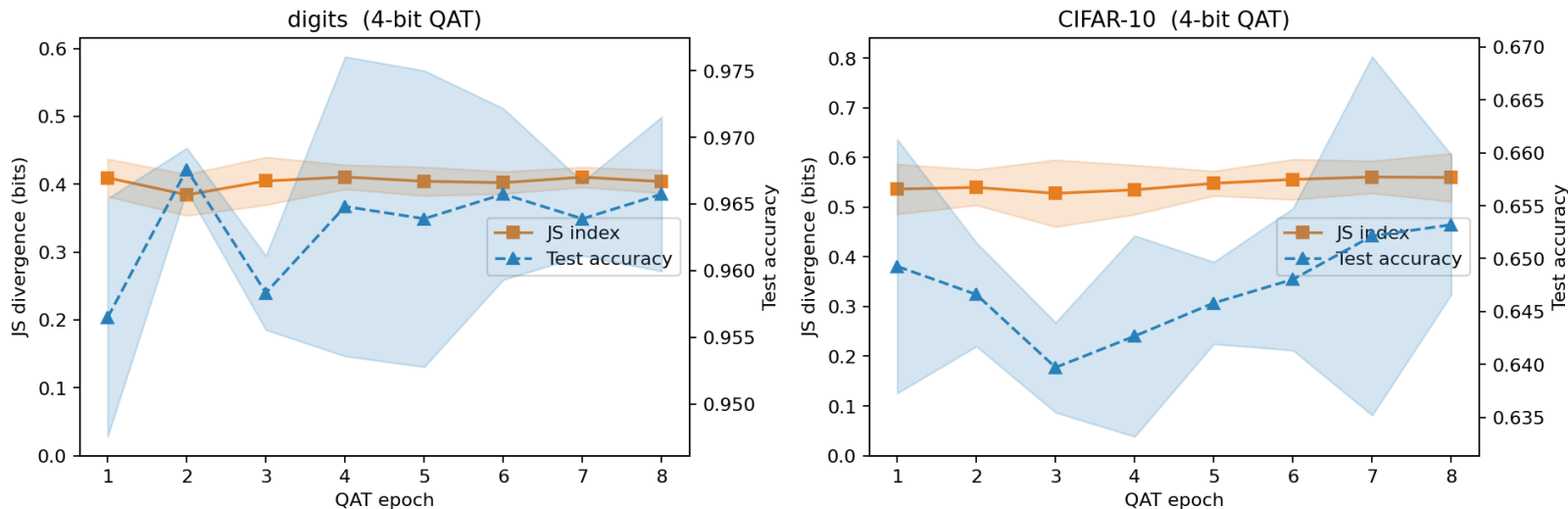
Statistical similarity index across datasets (3 seeds, mean $\pm$ std)



The index tracks quantisation severity almost perfectly ($\rho \approx -0.99$) and predicts the accuracy drop across three datasets of varying difficulty.

Dynamics during QAT

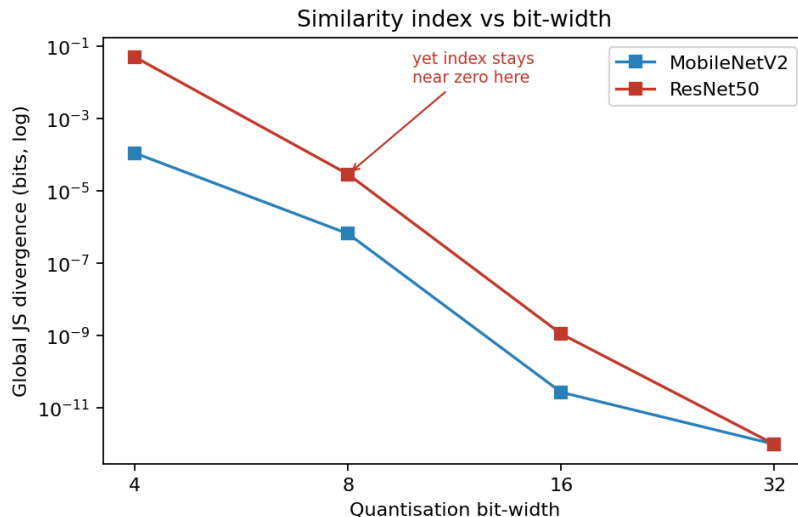
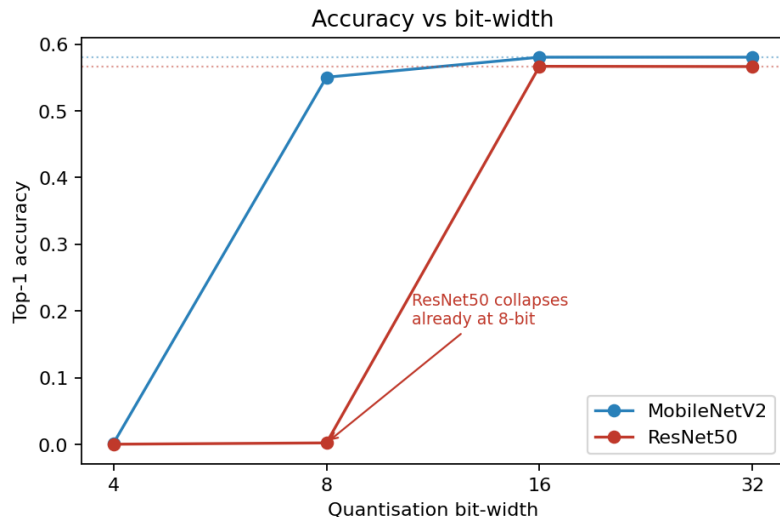
In-training dynamics during real 4-bit QAT (3 seeds, mean $\pm$ std)



During QAT the index holds steady at the level of the given bit-width (**weights remain genuinely quantised**), while accuracy is maintained or recovered against naive PTQ.

Generalisation to ImageNet

ImageNet (imagenet_v2 subset): accuracy collapse vs weight-distribution index



MobileNetV2 confirms the pattern; ResNet50 revealed the fragility of naive per-tensor quantisation in deep networks — **addressed by per-channel quantisation.**

Findings and limitations



Findings

- The index works as both a predictor (before training) and a monitor (during training).
- Dataset-agnostic — validated on three datasets.
- Independent of the chosen metric (KL/JS/cosine $|p| \approx 0.99$).
- Integrated directly into the training strategy.



Limitations

- Smaller models and a limited number of initialisations.
- Per-channel quantisation is necessary for deep networks.
- The global index misses activation-driven degradation (ResNet50 at 8-bit).

→ *motivates per-layer and activation monitoring.*

Next steps and discussion



Completion of WP3 + WP4 outputs and the project's final report (D17).



Future work: per-layer and activation monitoring, principled mixed precision.



Larger models, more initialisations, real int8 / TFLite export.

Discussion

Thank you for your attention.

